



Supporting Online Material for

Resource Partitioning and Sympatric Differentiation Among Closely Related Bacterioplankton

Dana E. Hunt, Lawrence A. David, Dirk Gevers, Sarah P. Preheim, Eric J. Alm,* Martin F. Polz*

*To whom correspondence should be addressed. E-mail: ejalm@mit.edu (E.J.A.); mpolz@mit.edu (M.F.P.)

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This PDF file includes:

Materials and Methods
Figs. S1 to S10
Tables S1 and S2
References

SUPPLEMENTAL ONLINE MATERIAL

TABLE OF CONTENTS

	Page
I. MATERIALS AND METHODS	2
II. SUPPLEMENTAL REFERENCES	10
III. WEBLINK TO AdaptML APPLICATION	11
IV. SUPPLEMENTAL TABLE AND FIGURES	11

I. MATERIALS AND METHODS

Sampling rationale

To investigate partitioning of *Vibrionaceae* strains in the water column, we examined their distribution among the free-living and associated (with particles and zooplankton) fractions of the bacterioplankton community at two time points. This was achieved by sequential filtration with decreasing pore size cutoffs and subsequent cultivation on *Vibrio* selective media (Fig. S1B). Here, we give additional details on sampling protocols and rationale supplementing the overview given in the main text.

Filtration is commonly used in oceanography to separate particle-associated and free-living populations by retention of particles on filters, although the filter size cut off for collecting particle-attached bacteria has varied in past studies between 0.8 and 10 μm (1-4). To obtain higher ecological resolution, we used sequential filtration since alternate types of particulate organic matter and organisms (e.g., phytoplankton, zooplankton) will have distribution maxima in different size fractions thus enabling differentiation of associated bacterial genotypes.

We collected a total of four size fractions with different expected composition of particles and organisms (Fig. S1B). The largest fraction ($\geq 63 \mu\text{m}$) was visually enriched in zooplankton and detrital material (e.g., pieces of macroalgae, terrestrial plant material); however, large gelatinous material [frequently part of marine snow, which represents particles $>0.5 \text{ mm}$ (5)] was likely not collected since it is disrupted by the pressure on the plankton nets used for collection. All other fractions were collected by gravity rather than vacuum filtration to minimize disruption of fragile particles. The large particle fraction (63-5 μm) likely contains zooplankton fecal pellets, dead and living algae, and other detritus. The composition of the 5-1 μm size fraction is somewhat ambiguous since it may contain both cells attached to very small particles as well as large or dividing cells; however, it provides a firm buffer between obviously particle-attached ($>5 \mu\text{m}$) and free-living ($<1 \mu\text{m}$) cells. Particulate material in this size range may include small algae, bacterial cell walls, as well as fragments of larger particles, which have broken apart; nonetheless, the small size of such particles is unlikely to sustain a resident bacterial population. Free-living bacteria, observed in the 1-0.22 μm size fraction, likely live on dissolved organic matter produced by living algae, cell lysis and the dissolution of particles.

Sample collection

Coastal ocean water samples were collected at high tide on the marine end of the Plum Island Estuary (NE Massachusetts) (Fig S1A) on two days representing spring (4/28/06) and fall (9/6/06) conditions in the coastal ocean. Nutrient concentrations, water temperature and chlorophyll levels were measured on both sampling dates (Table S1).

Two replicate samples of the largest size fraction (enriched in zooplankton) were collected by filtering $\sim 100 \text{ L}$ each through a 63 μm plankton net, which was subsequently washed with sterile seawater (Fig. S1B). Particle-associated and free-living bacterial populations were collected from quadruplicate water samples, which were independently

pre-filtered through the 63 μm plankton net (to remove the zooplankton-enriched fraction) into 4 L nalgene bottles (Fig. S1B). For each bottle, water was sequentially filtered through 5, 1 and 0.22 μm pore size filters, collecting at least four replicate filters per size fraction. To avoid disruption of fragile particles, the 63-5 and 5-1 μm fractions were collected on polycarbonate membrane filters (Sterlitech) using gravity filtration followed by washing with 5 ml of sterile (0.22 μm -filtered and Tindalized) seawater to remove free-living bacteria that might have been retained on the filter. The <1 μm fraction containing free-living bacteria was collected on 0.22 μm Supor-200 filters (Pall) by applying gentle vacuum pressure.

After size fractionation, particles and zooplankton were broken up before plating (Fig. S1B). The zooplankton sample was homogenized using a tissue grinder (VWR Scientific) and vortexed for 20 minutes at low speed before concentration on 0.22 μm Supor-200 filters (Pall). These filters were then plated directly on selective media. Similarly, 5 μm and 1 μm filters were placed in 50 ml conical tubes with 50 ml sterile seawater and vortexed at low speed for 20 min to break up particles and detach bacteria from the filters. The supernatant was concentrated on 0.22 μm filters, and both the original and supernatant filters were placed directly on media to collect isolates.

Strain isolation and identification

Isolates were obtained from TCBS plates (Accumedia or Difco) with 2% NaCl since this media has been shown to yield good correspondence in phylogenetic groups of vibrios detected by quantitative PCR and isolation (6). After 2-3 days of growth, colonies were counted and re-streaked a total of three times alternately on Tryptic Soy Broth (TSB) (Difco) with 2% NaCl and on TCBS media.

For classification of strains by sequencing, purified isolates were grown in marine TSB broth overnight; DNA was extracted using either a tissue DNA kit (Qiagen) or Lyse-N-Go (Pierce). Following the rationale of multilocus sequence analysis (MLSA), housekeeping genes were used for further strain characterization since these are unlikely to be under environmental selection. The partial *hsp60* gene sequence was amplified for all isolates as described previously (7). For isolates with an *hsp60* sequence differing by more than 2% from an already characterized strain, the 16S rRNA gene was PCR amplified using primers 27F- 1492R and sequenced using the 27F primer (8). The 16S sequence was used to identify the organism using the RDP classifier (9) and BLAST (10). In cases where the *hsp60* gene either failed to amplify or the sequence diverged greatly from other vibrios, 16S rRNA gene sequencing confirmed that these isolates largely belonged to the genera *Pseudomonas*, *Shewanella*, *Pseudoalteromonas*, and *Agaravorans* (RDP Classifier) (9). We excluded non-*Vibrionaceae* strains from further analysis.

To confirm relationships for *V. splendidus*, the most highly represented group among isolates, an additional gene (*mdh*) was sequenced. The partial *mdh* gene was amplified using primers *mdh_mod.for* (5'- GAY CTD AGY CAY ATC CCW AC -3') and *mdh_mod.rev* (5'- GCT TCW ACM ACY TCD GTR CCY G -3') (S. Preheim, unpublished data). Two additional housekeeping gene sequences were obtained (*pgi*, *adk*) for select groups of strains, using *pgi.for* (5' –GAC CTW GGY CCW TAC ATG GT

- 3') and *pgi*.rev (5'-CMG CRC CRT GGA AGT TGT TRT-3') (S. Preheim, unpublished data) and *adk*.for (5'- GTA TTC CAC AAA TYT CTA CTG G-3') and *adk*.rev (5'- GCT TCT TTA CCG TAG TA- 3') (11). All of these genes were amplified using the following PCR conditions: 2 min at 94°C followed by 32 cycles of 1 min each at 94°C, 46° and 72°C with a final step of 6 min at 72°C. Most genes were sequenced at least twice using forward and reverse primers. All sequencing was performed at the Bay Paul Center at the Marine Biological Laboratory, Woods Hole MA.

Phylogenetic tree construction and representation

The partial *hsp60*, *mdh*, *adk*, and *pgi* gene sequences yielded unambiguous alignments of 541, 422, 372, 395 nucleotides, respectively. Phylogenetic relationships were reconstructed using PhyML v.2.4.4 (12) with following parameter settings: DNA substitution was modeled using the HKY parameter (13); the transition/transversion ratio was set to 4.0; PhyML estimated the proportion of invariable nucleotide sites; the gamma distribution parameter was set to 1.0; 4 gamma rate categories were used; a BIONJ tree was initially used; and, both tree topology and branch lengths were optimized by PhyML. Circular tree figures were drawn using the online iTOL software package (14). To prevent numerical instabilities in AdaptML's maximum likelihood computations, branches with zero length were assigned the minimal observed non-zero branch length: 0.001.

Empirical statistical testing

We employed empirical statistics to quantify evidence for differential environmental distribution of phylogenetic groups (Fig. S2). We first tested the overall association of phylogeny with our environmental data using a non-parametric parsimony-based metric. We assigned a different character to each of the environmental categories, and calculated the minimum number of character transitions needed to explain the data given the observed *hsp60* phylogeny. Although this test is likely to be overly conservative given the heterogeneous nature of our observed clusters, it nonetheless supported a highly significant correlation between phylogeny and both size fraction ($p < 10^{-49}$) and season ($p < 10^{-79}$). Exact p-values were computed based on the algorithm of (15). It was not possible to compute p-values for both season and size fraction together because the computational complexity of the algorithm grows exponentially with the number of character states.

We also employed non-parametric empirical statistics to test specific model predictions. We tested the hypothesis that each of the clusters identified by the model would be likely to arise by chance. To do this, we produced a 2 X 8 contingency table to test for any associations between cluster membership and distribution across environments. We used the Fisher exact test (16) as implemented in the R programming language to evaluate the significance of each association. The results are shown in Table S2.

Overview of AdaptML

We developed a maximum likelihood method to help identify the boundaries of ecologically distinct populations and infer the ancestral habitat association of internal nodes in the strain phylogeny. The key to our method is a hidden variable mapping a ‘projected habitat’ to each node. We mathematically characterize each habitat as a discrete probability distribution, which describes the likelihood that a strain adapted to that habitat will be observed in each of our eight environmental categories. These distributions, which we refer to as emission probabilities in accordance with terminology used in machine learning, are not known *a priori* and must be learned from the data. Because of this probabilistic definition of habitats, a phylogenetic group spanning several environmental categories can still be considered an ecologically distinct population.

Using the habitat variables and isolate sequence-based phylogeny, we built a hidden-Markov model (HMM) (16) describing the evolution of habitat association (Fig. S3). The probability that adjacent nodes in the phylogeny share the same habitat is a function of both the branch length separating them and a parameter that represents the rate at which a lineage can transition between habitats (the transition rate). The observed variables – the environmental category from which each strain was sampled – occur only at the leaves of the phylogeny. The parameters necessary for our model can be learned from the data according to the following algorithm:

1. Initialize parameters. We initialize 16 habitats, each with random emission probability distributions over the 8 environmental categories. The transition rate parameter is initialized to 10^{-1} transitions per substitution/site (relative to the gene phylogeny branch lengths).
2. Infer the observed data’s likelihood, given phylogeny and parameter estimates. We use a dynamic programming algorithm and the model parameters (transition rate and emission probabilities) to compute the likelihood of the observed data (environmental category for each isolate). Our computation proceeds in a manner identical to Felsenstein’s “pruning” method of computing likelihoods on a phylogenetic tree (17).
3. Optimize parameters to maximize likelihood of observed data. We estimate the probability that each internal node is associated with a given habitat by summing over all possible habitat assignments at other nodes (E-step). These probabilities are used (M-step) to update the:
 - a. *Transition rate parameter:* We numerically optimize the transition rate to maximize the likelihood of the observed data.
 - b. *Emission probability matrix:* We update the emission probability matrix by taking the matrix that maximizes the likelihood of the observed data given the marginal likelihoods for the habitat assignments at each of the phylogeny’s leaves.

We note that separating these two steps represents an approximation, as these two parameters are not strictly independent. The approximation, however, speeds up

the implementation considerably as only one parameter (instead of $1 + 16 \times 7 = 113$) is optimized numerically.

4. Test for convergence. If the model parameters do not change significantly from the previous iteration, then the emission probabilities and the transition rate are considered to have converged: continue to step (5). (A typical trajectory of emission probability convergence is shown in Fig. S4. Note: the approximation identified in step 3 can lead to fluctuation near a likelihood maximum rather than actual convergence). Otherwise, return to step (2).
5. Test for model complexity/redundancy. If a pair of habitats has emission probability distributions that exhibit correlations greater than 0.90, they are merged into a single habitat and the algorithm continues from step (2). If no habitats are merged, the parameter estimation loop terminates. Although our approach employed manual inspection and empirical testing rather than a likelihood-based criterion for reducing model complexity [such as the AIC (18)], our algorithm can be easily extended to include a likelihood criterion. To test for overfitting, we performed simulations as described in the main text and Figure S6. We found that our scheme acts conservatively since it usually underestimated the true number of habitats. Figure S9 shows how the inferred habitats identified by the model vary with different cutoffs.

Once a set of model parameters has been learned, we utilize the following protocol to identify ecologically distinct and statistically significant populations.

1. Infer node habitat assignments that maximize the 'joint' probability of the observed data. We rely upon a joint likelihood calculation to infer a single habitat assignment per ancestral node. To compute this likelihood, we use the parameter estimates inferred by the algorithm described above, which sums over all habitat assignments. Phylogenetic groups that share a common habitat association are taken as candidate ecological populations if they pass an empirical significance test.
2. Empirical testing to identify ecologically distinct populations. The assignment of nodes to habitats in step (1) identifies the most likely set of population boundaries, but may include some weakly predicted clusters. To filter low confidence ecological groupings, we estimate empirical p-values for each clade and only report statistically significant ($p < 0.0001$) populations (Fig. 1, A and B). Empirical p-values are computed by comparing the likelihood of the parent node for a cluster to the likelihood observed at the same node in randomized trials where environmental assignments are shuffled, but phylogeny is maintained. For comparison, all possible clusters (with no significance cutoff) can be inferred from the full model results shown in Fig. S7. Finally, we report results from non-parametric empirical testing for each of the clusters in Table S2 (as described above under "Empirical statistical testing").

Detailed description of maximum likelihood model

Conditional Likelihoods:

The conditional likelihood describes the likelihood L that the leaves of a subtree exhibit their observed states, conditional on the subtree's root node k taking state s . This likelihood can be defined recursively, assuming two child nodes l and m

$$L_k(s) = \left(\sum_x P(x | s, t_l) L_{kl}(x) \right) \times \left(\sum_y P(y | s, t_m) L_{km}(y) \right)$$

where the function $P(x|s,t)$ represents the probability of transitioning between states x and s along some interval t . To reduce the number of fitted parameters early in our model, we use the simplifying assumption that all state transitions can be described using the same transition rate parameter μ . Thus, we compute the probability of transitions between states as

$$P(x | y, t) = \frac{1}{h} (1 - e^{-h\mu t})$$

and the probability of remaining in the same state

$$P(x | x, t) = \frac{1}{h} (1 + (h-1) e^{-h\mu t})$$

which is analogous to the Jukes-Cantor model for nucleotide sequence evolution (16). Note that if k is a leaf node in state s with observed environment o , the likelihood is drawn from the emission probability matrix P_e :

$$L_k(s) = P_e(o | s)$$

Marginal and Joint Likelihoods:

To compute the marginal likelihood that a node (k) has state s , we combine the conditional likelihoods:

$$ML_k(s) = \frac{1}{K} \times \prod_l \left[\sum_x P(x | s, t_{kl}) L_l(x) \right]$$

where the l are drawn from the set of nodes adjacent to k , and K is a normalization factor such that $\sum_s ML_k(s) = 1$.

To compute the likelihood of the observed data for the single best assignment of habitats to nodes (17), the summation terms in the likelihood formula are replaced with **max** operations. This is equivalent to maximizing the ‘joint’ likelihood:

$$L_k(s) = \left(\max_x P(x | s, t_{kl}) L_l(x) \right) \times \left(\max_y P(y | s, t_{km}) L_m(y) \right)$$

The backtracking process used to keep track of the **max** arguments is analogous to the Viterbi algorithm for finding the most probable state path in an HMM (16).

Parameter estimation:

As probability distributions, each set of emission probabilities must satisfy

$$\sum_o P_e(o | s) = 1$$

Because leaf nodes are independent samples of their emission probability distributions (given their state assignment), we use a weighted-average approach to calculating the emission probability matrix P_e

$$P_e(o | s) = \frac{1}{K} \times \sum_k ML_k(s) \times \Delta(o, o_k)$$

where $ML_k(s)$ is the marginal likelihood that node k is adapted to habitat s , the function $\Delta(x,y)$ equals 1 if and only if x equals y (and is 0 otherwise), and K is a normalization factor.

We learn the transition rate parameter μ by numerical optimization to maximize the likelihood of the observed data (summed over all values for the hidden variables).

Defining ecologically coherent and statistically significant clusters

We use empirical testing to estimate the statistical significance associated with the likelihood value computed for each node in the maximum (joint) likelihood assignment of habitats to nodes (given the final parameter estimates). Each trial preserved the phylogeny, the inferred habitat assignments, and the habitat and transition rate parameters, but environmental categories at the leaves were shuffled. The maximum 'joint' likelihoods at each node were compiled over all trials and used as empirical background probability distributions.

We use empirical testing to estimate the statistical significance associated with the likelihood value computed for each node in the maximum (joint) likelihood assignment of habitats to nodes (given the final parameter estimates). Each trial preserved the phylogeny, the inferred habitat assignments, the habitat and transition rate parameters, and the frequency of the various environmental categories; however, each leaf was randomly assigned an environmental category in each trial. The maximum 'joint' likelihoods at each node were compiled over all trials and used as empirical background probability distributions.

Using the habitat associations learned from the original (non-randomized) data set, we identified internal nodes where habitat transitions were inferred to take place. These nodes were then iterated through using a post-fix traversal:

Nested Clusters:

At each of these "transitional" nodes, a second, pre-fix traversal took place. If the subtree rooted by the current node possessed a likelihood greater than that observed in 99.99% of the random trials and 90% of its leaves shared the current node's habitat assignment, we recognized the subtree as an ecologically coherent and statistically significant cluster. This latter cutoff (90% coherence) was necessary to ensure meaningful clusters because the likelihood at a parent node can be unusually high when two nearly significant (but non-identical) child groups are combined. Requiring a majority of child nodes to have the same predicted model state as the parent has the effect of identifying clusters that correspond more directly to single putative 'populations'. Making the

threshold too high (*e.g.*, 100%) would eliminate some larger clusters that have a small, internal nested cluster.

To enable the discovery of nested clusters, identified clusters were pruned from the phylogeny. Nodes representing clades that contained the cluster had their joint-likelihoods modified so that they no longer incorporated information from the pruned groups. If the clade rooted by the current node did not satisfy both the p-value and 90% coherence thresholds, the second recursion would descend to the current node's children. The second recursion only terminated if the current node was either a leaf, itself a habitat transition node, or determined to root an ecologically coherent and statistically significant cluster.

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III. WEBLINK TO AdaptML APPLICATION

An online version of AdaptML can be found at: <http://www.almlab.org/adaptml/>. Here, users can upload their own phylogenetic trees and sampling metadata for automated analysis. Copies of the AdaptML source code can be found at this address as well.

IV. SUPPLEMENTAL TABLE AND FIGURES

This section contains Supplementary Table S1 & S2 and Supplementary Figures S1 to S10. See main text for details.

Table S1. Temperature and nutrient concentrations on sampling dates.

	Temperature	Chlorophyll a ¹	DOC ²	TDN ²	NO ₃ ⁻ +NO ₂ ⁻	NH ₄ ⁺	TDP ²	PO ₄ ³⁻
	[°C]	[µg/L]	[mg C/L]	[mg N/L]	[µg N/L]	[µg N/L]	[µg P/L]	[µg P/L]
Spring (4/28/06)	11	4.07	2.11	0.17	9	189	18	14
Fall (9/6/06)	16	6.03	2.28	0.27	5	144	24	25

¹ measured using overnight extraction in 90% acetone (19)

² DOC = dissolved organic carbon, TDN = Total Dissolved Nitrogen, TDP = total dissolved phosphorous. All chemical analyses were performed at the University of New Hampshire, Durham, NH.

Table S2. Empirical significance testing of ecologically differentiated clusters predicted by the model. Fisher’s exact test was used to identify significant associations between clusters and environments as described in the SOM. Group numbers correspond to Figure 1 in the main text.

Group	P-value
1	1.16 E-07
2	1.04 E-16
3	5.06 E-15
4	1.70 E-01
5	1.11 E-02
6	1.26 E-04
7	7.54 E-10
8	2.46 E-05
9	2.35 E-07
10	7.19 E-07
11	1.26 E-13
12	2.62 E-03
13	5.13 E-10
14	1.16 E-06
15	1.01 E-02
16	1.13 E-08
17	4.03 E-07
18	2.81 E-03
19	1.08 E-08
20	1.80 E-06
21	2.31 E-13
22	7.02 E-03
23	3.32 E-06
24	6.06 E-14
25	2.76 E-03

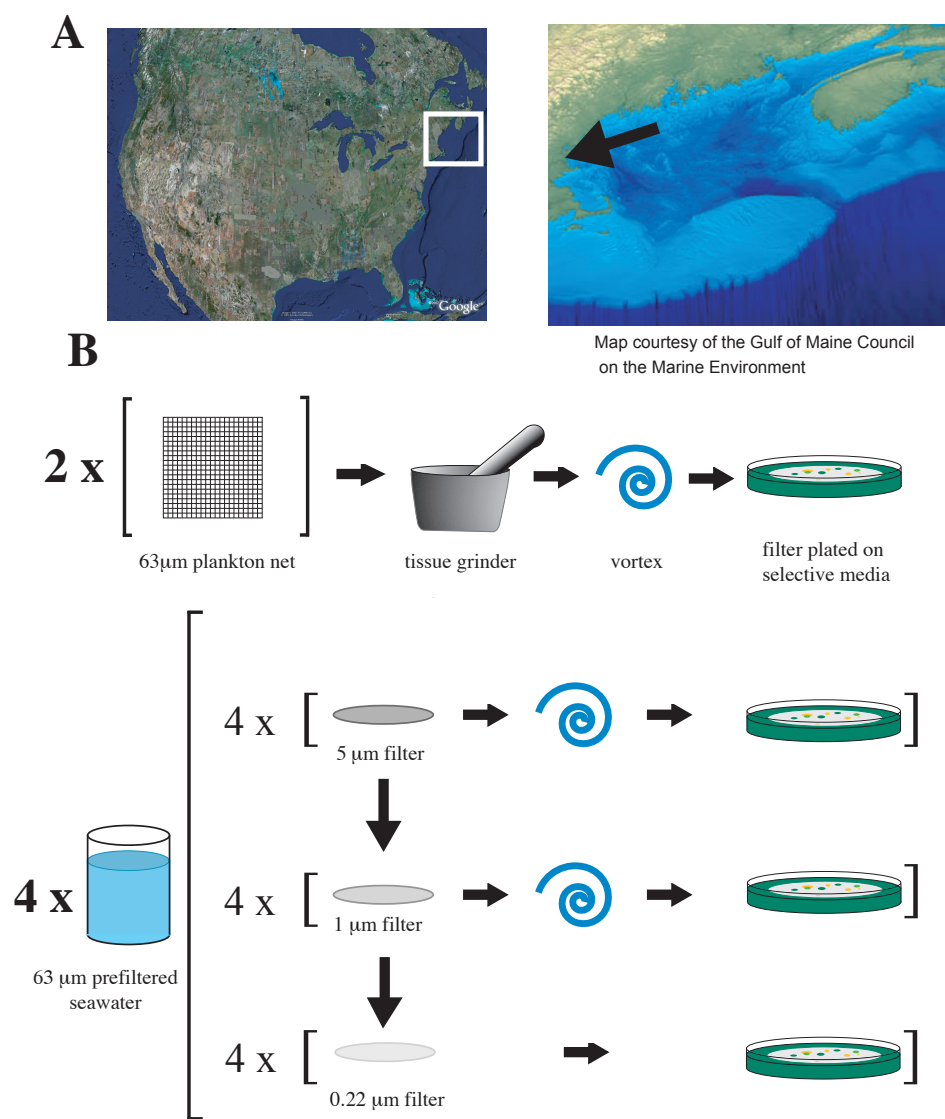


Fig. S1. Sampling site and outline of sampling strategy for determination of bacterial distribution among seasons and size fractions in coastal water. **(A)** Sampling location shown on a map of North America (left) with a white box depicting the bounds of the picture at right: the Gulf of Maine. The arrow points to the sampling location, Plum Island Sound, MA. **(B)** Protocol for obtaining size fractionated of bacterial seawater isolates using sequential filtration and plating on selective media.

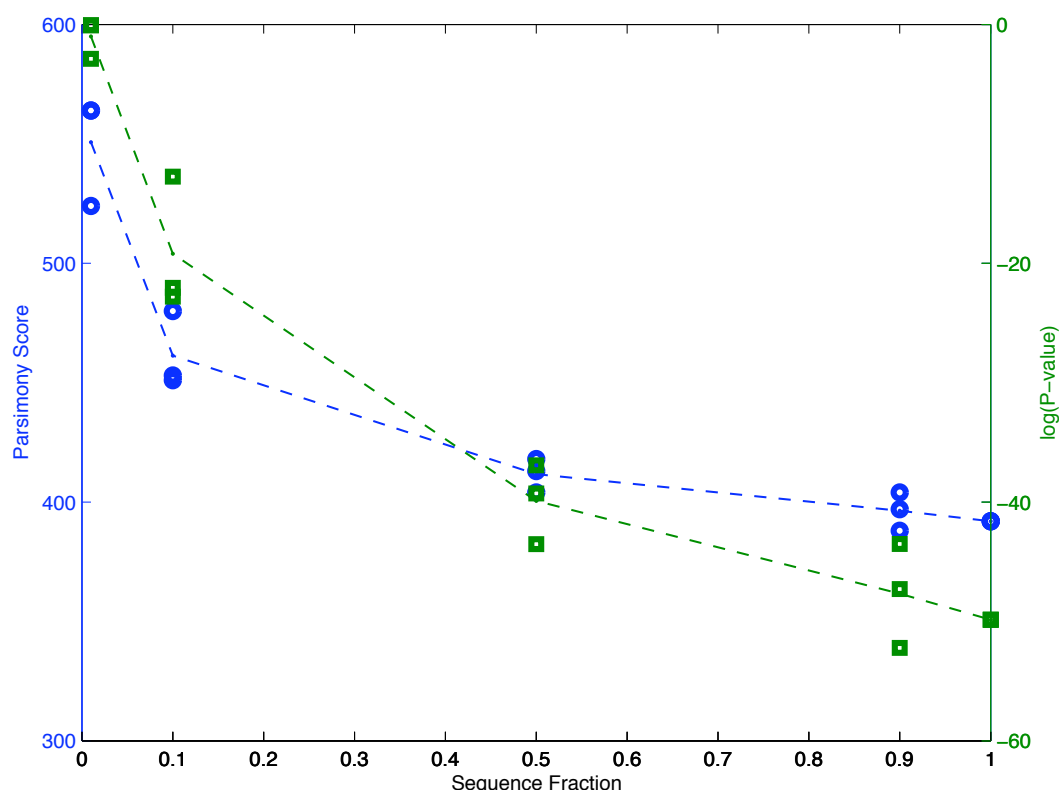


Fig. S2. Empirical statistical testing for ecological association of phylogenetic clades. The likelihood that the observed parsimony score (or lower) for size fraction data might have arisen by chance was calculated [using the method of (15)] for a series of trees, inferred using subsets of strains from the *hsp60* gene alignment. To test the effect of statistical uncertainty on the inferred association, 1%, 10%, 50%, 90%, or 100% of the sequence data was randomly selected and used to construct the phylogeny (with 3 replicates each). The parsimony score for each such tree is shown with green dots corresponding to the left axis; the p-value of obtaining that parsimony score by chance is shown with blue dots corresponding to the right axis. These results are based on size fraction data only, but similar results are obtained with season data.

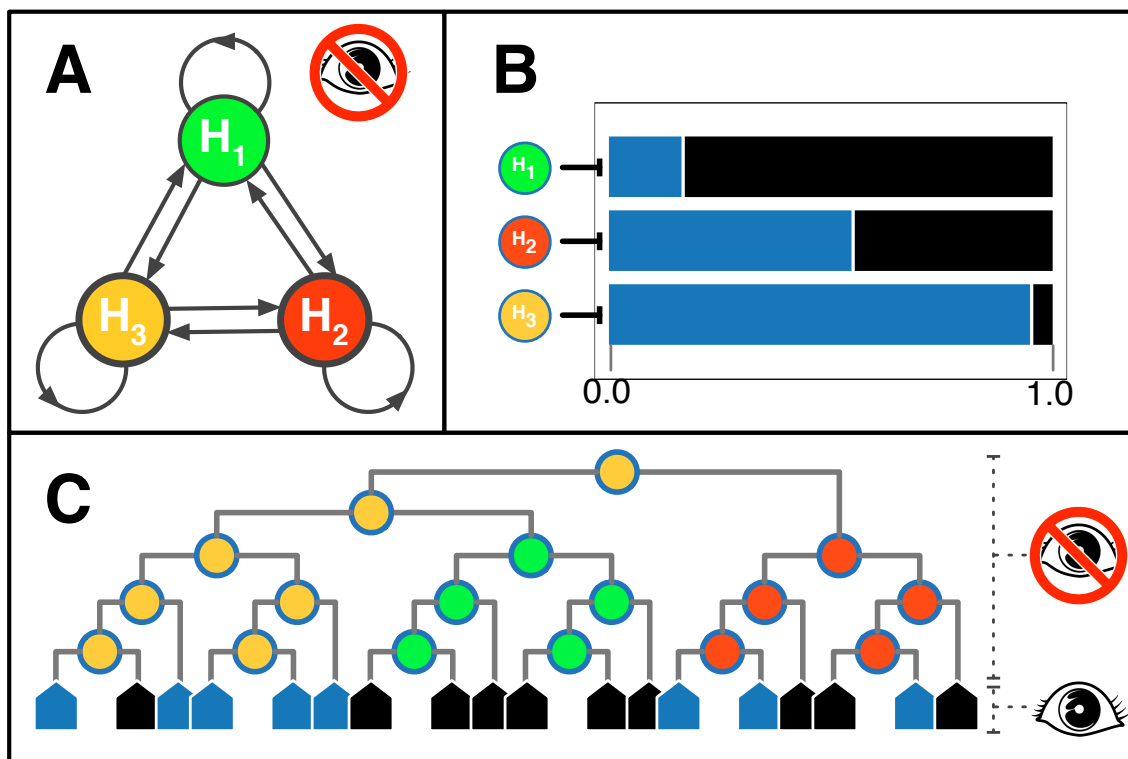


Fig. S3. Overview of hidden Markov model components. **(A)** Habitats represent hidden (latent) variables in the model; experiments do not directly measure what habitat a strain is adapted to. Two adjacent nodes on the phylogeny may differ in their habitat assignment according to the rate of habitat transition (arrows between habitats). In our simple model, all transition rates are equivalent. **(B)** Associated with each habitat is an emission probability distribution describing how likely a strain associated with a particular habitat (H_{1-3}) is sampled from a given environment (blue or black). Bars in this cartoon depict hypothetical probability distributions; strains adapted to Habitat 1 have higher probability of being observed in the black environment than in the blue environment. **(C)** Our model maps a habitat onto each node in the phylogeny such that it maximizes the probability of the observed data (at the leaves). Observations are limited to the leaves of the phylogeny, as we cannot directly sample ancestral strains. In the example shown, the mostly blue group is mapped to habitat 3, the mostly black group to habitat 1, and the heterogeneous group is mapped to habitat 2.

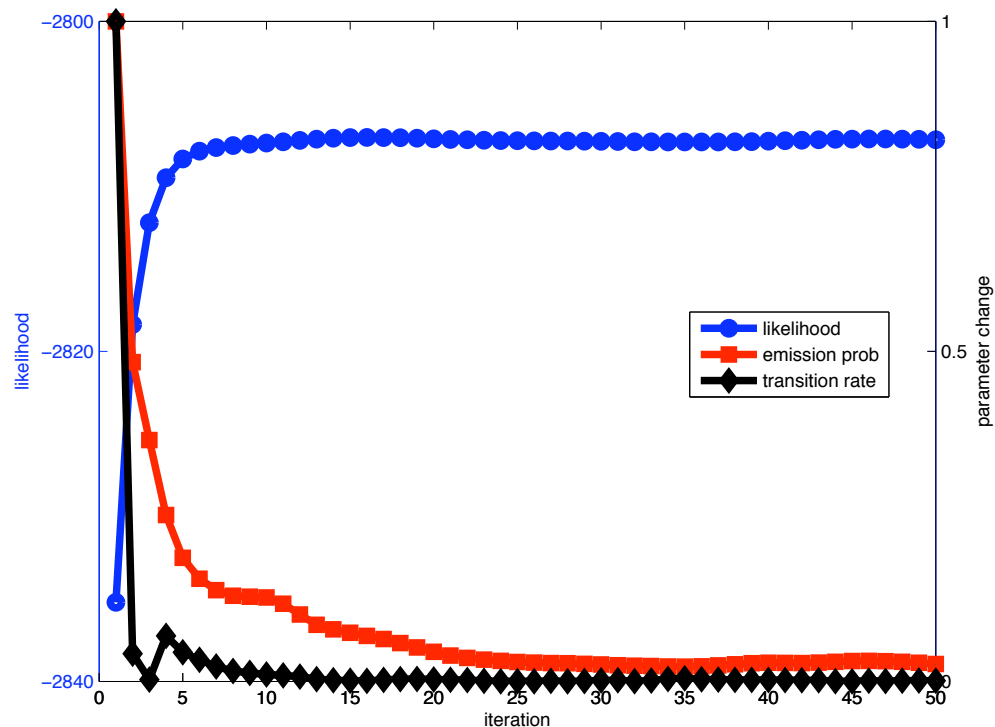


Fig. S4. Convergence of iterative parameter optimization. During the parameter optimization, both the average change in probability for components of the emission probability matrix (red line) and the change in transition rate (black line) decrease rapidly. The overall log-likelihood of the observed data converges in concert with the parameter estimates (blue line).

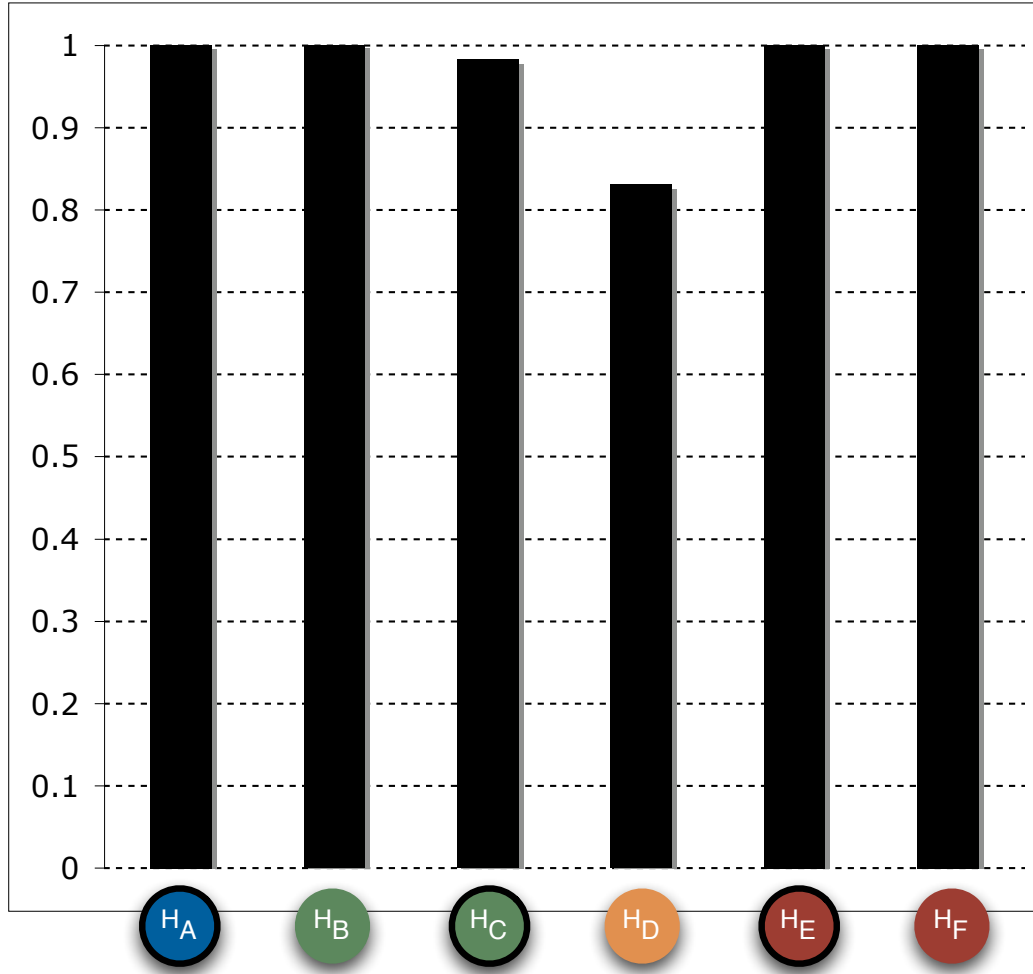


Fig. S5. Reproducibility of inferred habitats. Sixty independent trials of the iterative habitat learning process were performed. Shown are the frequencies of occurrence of the habitats presented in Figure 1. The habitat similarity cut-off for counting a match was an average emission probability difference < 0.10 over all environmental categories. As expected, the least abundant habitat in our data, H_D (see Fig. S4) is the least reproducible, although even this habitat is identified in 83% of trials.

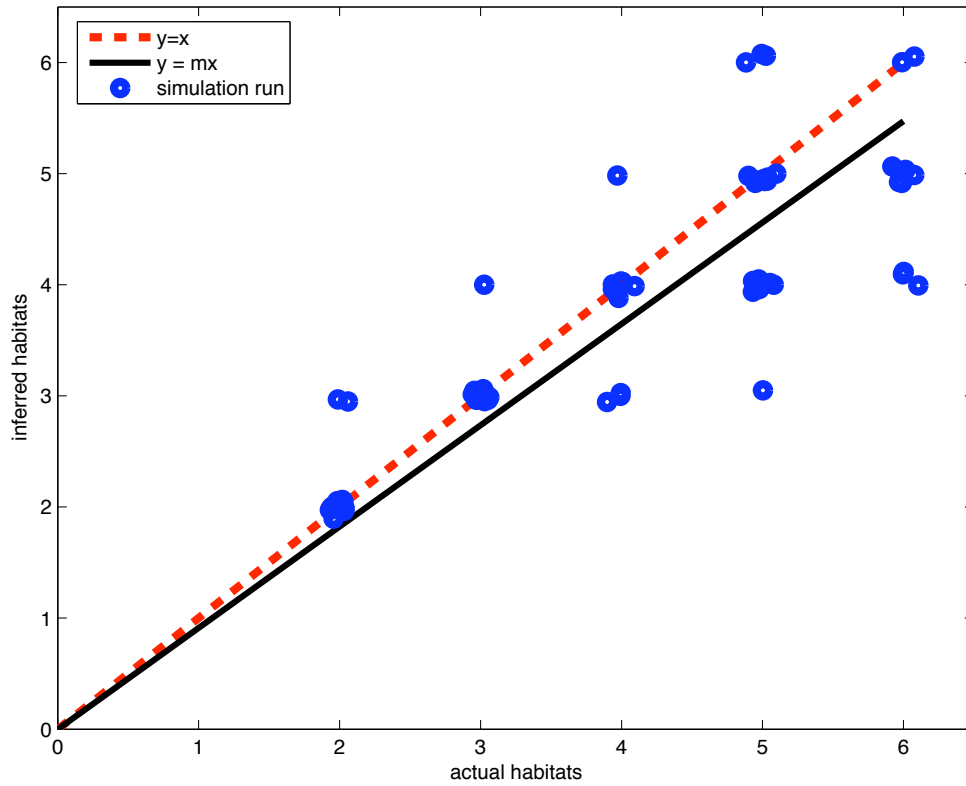


Fig. S6. Number of habitats inferred from simulated data. We generated 120 simulated datasets and compared the number of inferred habitats to the true number. Each dataset was randomly assigned between 2 and 6 habitats; these habitats were distributed over environmental categories by randomly partitioning the interval (0,1) according to a uniform distribution, but requiring that each successive partitioning must occur at a larger value than the last (random partitioning without this restriction leads to a large number of similar ‘generalist’ habitats). These habitats were mapped onto the set of clusters learned from the *Vibrionaceae* data (Fig. 1, main text). Shown are the number of habitats inferred for these trials *versus* the number actually present (a small amount of Gaussian noise was added to each data point so that the data points could be discerned). These results suggest that the habitat inference algorithm is generally conservative, inferring fewer rather than more than the true number of habitats.

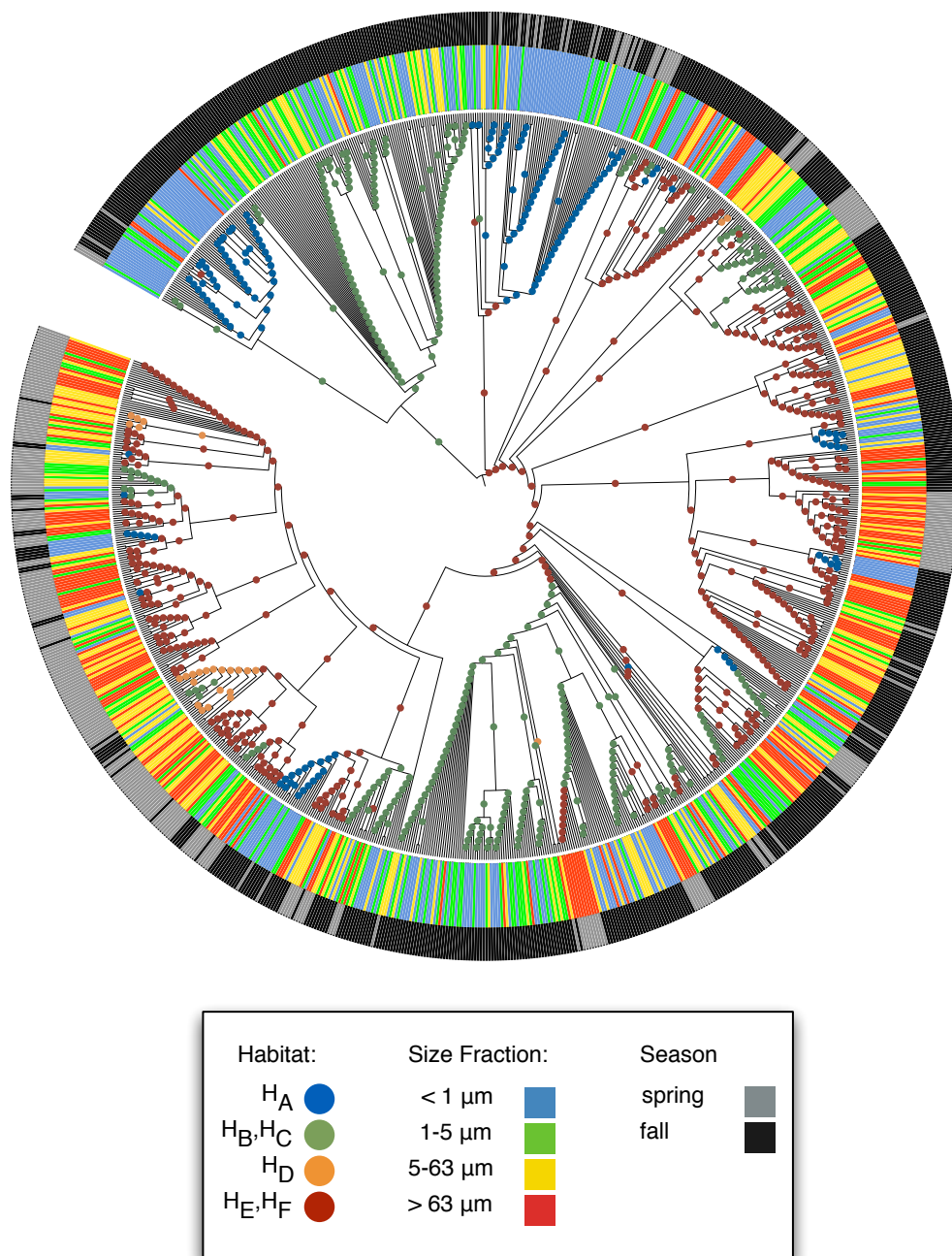


Fig. S7. Inferred habitat associations for all ancestors of sequenced *Vibrio* strains. The rings surrounding the tree represent the season (outer) and size fraction (inner) from which strains were isolated. The maximum likelihood assignment of nodes to habitats is shown for all nodes, regardless of the confidence of each prediction (only confident assignments are shown in main text Fig. 1A). Colored circles on each branch indicate the habitat assignment (H_{A-F}, as in Fig. 1 main text) for the node immediately below that branch (see above legend for color scheme). Branch lengths are adjusted to aid visualization and do not represent evolutionary distances.

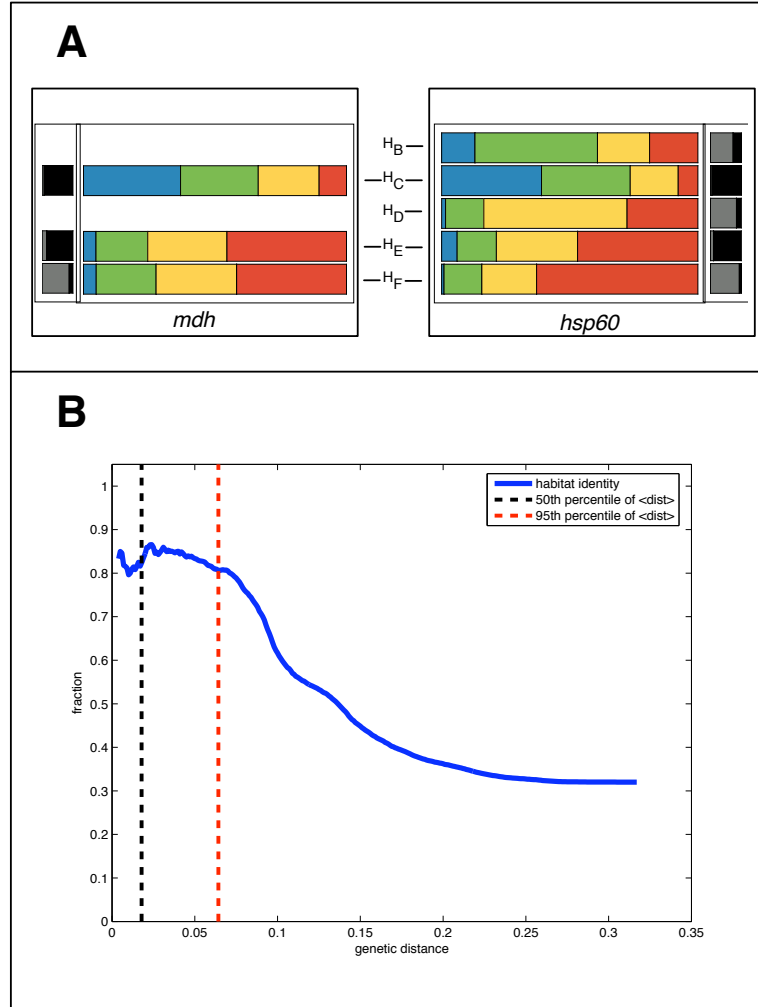


Fig. S8. Comparison of habitat inference on different gene phylogenies (*hsp60* and *mdh*) for *Vibrio splendidus* strains. **(A)** Juxtaposition of habitats learned from the *mdh* and *hsp60* datasets for *V. splendidus* strains only; habitats are labeled to allow comparison with habitats predicted for all *Vibrionaceae* in Fig. 1, main text. Emission probabilities are normalized by the total number of isolates obtained in each environmental sample to reduce the effects of sampling bias. As expected, fewer habitats are identified from the *mdh* phylogeny, which has a lower rate of nucleotide divergence and thus is less well-resolved. **(B)** Comparison of habitat assignments to nodes. Because it is difficult to map the internal nodes between topologically distinct trees, the habitat assignment for the last common ancestor of each pair of *V. splendidus* strains was compared. If both corresponded to the “same” habitat (H_C, H_E, or H_F in both phylogenies), they were considered to be in agreement, otherwise they were considered to be in disagreement. The fraction of nodes in agreement is shown as a function of increasing genetic distance between the pairs of strains considered (in the *hsp60* phylogeny). The black and red lines indicate distances that include 50% and 95% of strains within the same cluster in main text Figure 1, respectively.

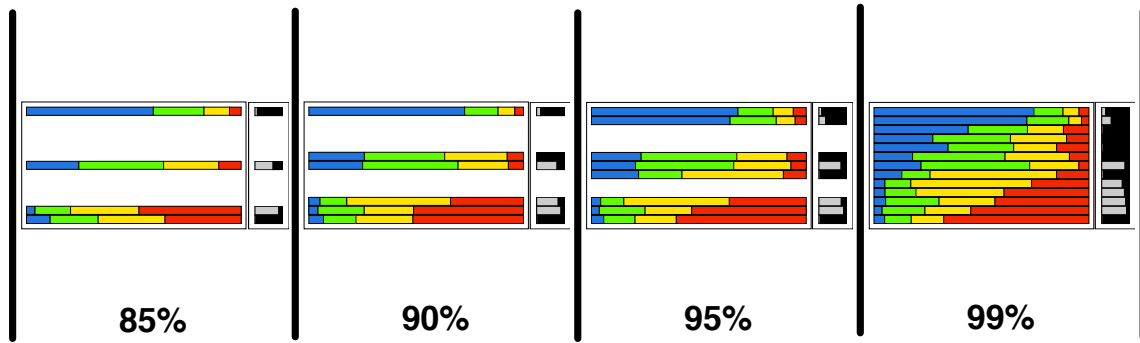


Fig. S9. Influence of the model complexity/redundancy parameter on inferred habitats. Clusters are merged during the model fitting procedure when the vectors describing their distribution across environments are more than 90% correlated. As this cutoff is varied, slightly different habitats are observed. At 85%, habitat H_C is not recovered; at higher values additional habitats become more redundant suggesting that the 90% cutoff allows conservative recovery of characteristic projected habitats.

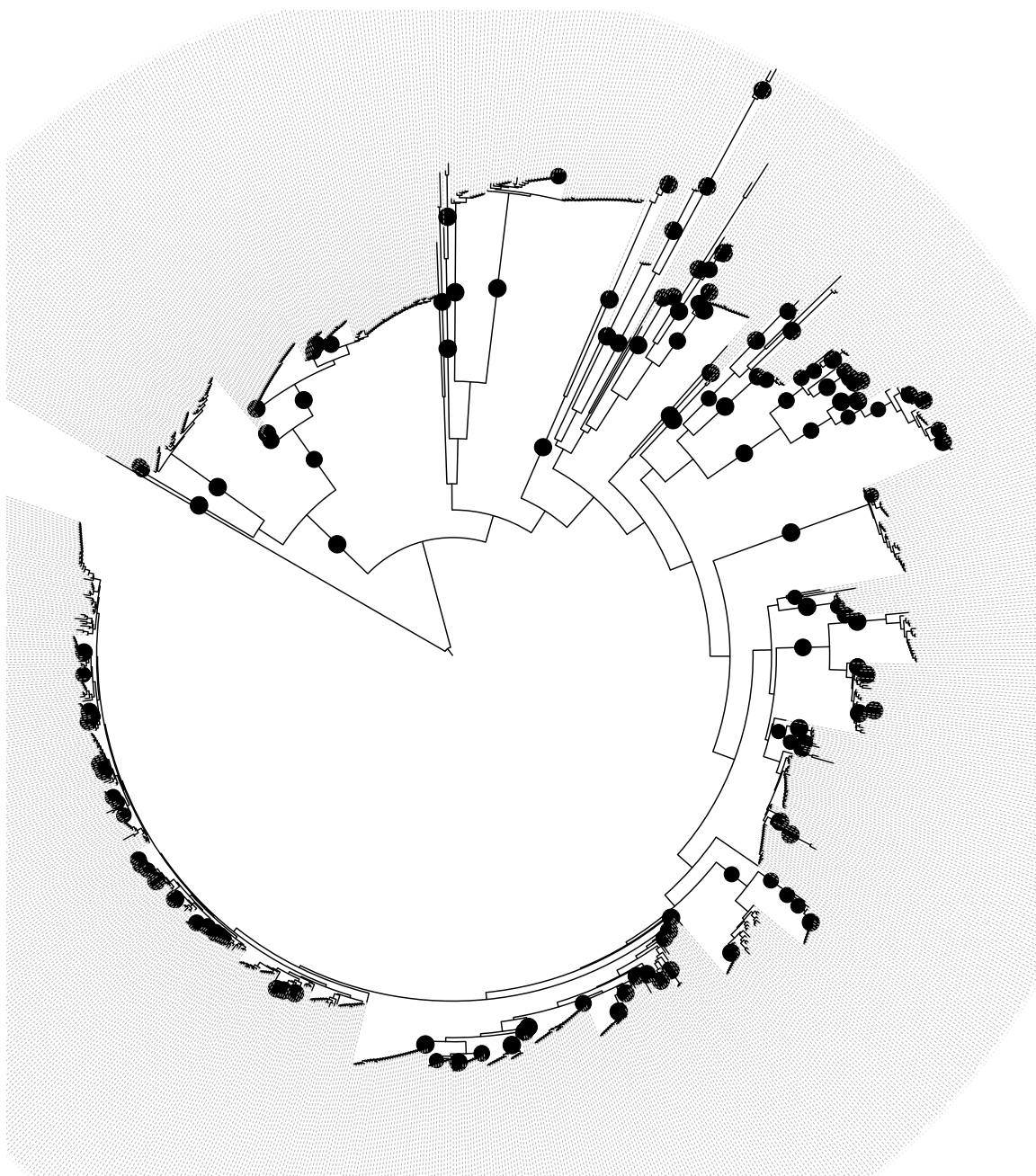


Fig. S10. Statistical uncertainty in the *hsp60* gene tree by constructing 100 trees using non-parametric bootstrap re-sampling from the *hsp60* alignment. Clades supported in greater than 80% of bootstraps are indicated with a black dot.